

Swetha Singam

Senior DevOps Engineer

☎ +91-8951009672 🔗 [linkedin.com/in/swethasingam232](https://www.linkedin.com/in/swethasingam232) ✉ singam.swetha.ramireddy@gmail.com

Professional Summary

Senior DevOps Engineer with 5+ years of experience in designing, automating, and managing scalable cloud infrastructure across AWS, GCP, and Azure. Expertise in Kubernetes, Terraform, CI/CD pipelines, and Infrastructure as Code (IaC). Strong focus on cloud cost optimization (FinOps), system reliability (SRE), and security.

Areas of Expertise

- Kubernetes & Container Orchestration
- Infrastructure as Code
- CI/CD automation
- Cloud Security & IAM
- FinOps
- Monitoring & Observability
- Production Support & Incident Management
- Multi-Cloud Architecture
- Cost Optimization
- Release Management
- MLOps

Projects - PoCs

SENTRA - Smart Engine for Neural Training, Registry & Automation

- Architected an end-to-end MLOps automation of ML lifecycle including training, validation, deployment, monitoring, & retraining.
- Designed ML CI/CD pipelines integrating MLflow, Databricks Jobs API, and GitHub for reproducibility, governance, and version-controlled deployments.
- Implemented model versioning, registry, and experiment tracking using MLflow with lineage tracking and hyperparameter tuning.
- Developed Kubernetes orchestration pipelines reducing model retraining time from days to hours through workflow automation.
- Engineered feature engineering and data pipelines (ETL/ELT) supporting batch processing and scalable data ingestion.
- Implemented model monitoring with data and concept drift detection, enabling continuous performance validation.
- Automated retraining pipelines triggered by drift and performance degradation, improving model accuracy and efficiency.
- Containerized ML workloads using Docker and integrated Prometheus/Grafana for observability, alerting, and scalable deployments.

TARA - Trusted, Accurate & Responsive Assistant

- Designed and implemented an LLMOps platform for Generative AI, enabling scalable orchestration of LLM pipelines and workflows.
- Built end-to-end ML pipelines using Kubeflow with workflow automation for training, evaluation, and deployment.
- Architected a production-grade RAG system using embeddings and vector databases for context-aware retrieval and responses.
- Implemented prompt engineering, prompt tuning, and model evaluation pipelines to improve LLM accuracy and performance.
- Integrated LangChain and LlamaIndex for modular orchestration across retrieval, inference, and post-processing layers.
- Leveraged OpenAI and Hugging Face models for scalable LLM inference and experimentation.
- Developed FastAPI-based inference services supporting real-time and batch predictions with low latency.
- Enabled observability with Grafana, including monitoring, logging, tracing, alerting, and governance via artifact versioning.

Professional Experience

Senior DevOps Engineer, (Infoservices)

Hyderabad Sep/2023 - Present

- Architected scalable GCP infrastructure using Terraform/Terragrunt across GKE, VPC, IAM, BigQuery, and Cloud Storage, reducing provisioning effort by 40%.
- Managed GKE clusters with autoscaling, RBAC, Helm deployments, and pod affinity across dev, QA, & production environments.
- Implemented GitOps workflows using Atlantis, ArgoCD, and Terrateam, reducing deployment time from 4 hours to 15 minutes.
- Built reusable Terraform modules standardizing GCP infrastructure provisioning and eliminating configuration drifts.
- Developed Jenkins and GitHub Actions CI/CD pipelines improving deployment reliability by 30% and accelerating release cycles.
- Deployed Dockerized microservices on GKE with autoscaling and zero-downtime deployment for high-availability workloads.
- Engineered Python and BigQuery FinOps dashboards identifying \$120K annual savings by cloud rightsizing & utilization analysis.
- Strengthened cloud security using IAM, RBAC, secrets manager, & automated credential rotation reducing manual effort by 50%.
- Implemented Prometheus, Grafana, Loki, and GCP Monitoring solutions reducing MTTR by 40% through centralized observability.
- Automated Kubernetes operations using Python and Bash scripts for cluster maintenance, monitoring, and compliance reporting.
- Designed secure hybrid cloud networking with shared VPCs, firewall policies, load balancing, and private connectivity solutions.
- Collaborated with SRE and engineering teams to improve deployment stability, release governance, and operational reliability.

DevOps Engineer, (Increff)

Bengaluru Nov/2022 - Aug/2023

- Engineered scalable GCP infrastructure using GKE, Compute Engine, Cloud IAM, Load Balancing, and Cloud Storage services.
- Managed GKE clusters supporting containerized workloads with autoscaling, rolling updates, and Helm-based deployments.
- Automated infrastructure provisioning using Terraform and Ansible reducing manual setup effort by 50% across environments.
- Developed Jenkins CI/CD pipelines integrating Maven, Docker, SonarQube, and automated cloud-native deployments.
- Built optimized Docker multi-stage container images for scalable deployments across dev, staging, and production environments.
- Configured GCP Monitoring, Logging, Alerting, and Grafana dashboards reducing incident detection time by 40%.
- Designed secure VPC architectures with firewall rules, load balancing, and hybrid connectivity for high-availability applications.
- Implemented IAM governance and automated credential lifecycle management improving cloud security and compliance readiness.
- Developed BigQuery FinOps dashboards delivering 15–20% infrastructure optimization through proactive cost analysis.
- Supported production operations, troubleshooting, and release deployments improving reliability across cloud-native platforms.
- Developed Python and Bash automation scripts for cloud operations, infrastructure management, and reporting workflows.

Application Development Associate, (Accenture)

Hyderabad Dec/2021 - Nov/2022

- Built Azure DevOps and Jenkins CI/CD pipelines improving deployment efficiency by 20% and reducing release defects by 25%.
- Automated build and deployment workflows using Jenkins, Docker, Maven, and PowerShell reducing manual intervention by 40%.
- Provisioned Azure infrastructure using IaC across Virtual Machines, VNets, Load Balancers, Storage Accounts, and IAM services.
- Deployed containerized applications on Azure ensuring scalable and highly available cloud-native application delivery.
- Configured Azure networking including VNets, NSGs, and Load Balancers improving infrastructure reliability and security.
- Implemented Azure monitoring and logging solutions improving operational visibility and troubleshooting efficiency.
- Automated configuration management and deployments using Ansible and Azure DevOps release pipelines.
- Supported Git branching strategies, release governance, and deployment standardization across engineering teams.
- Resolved production and deployment issues reducing incident resolution time by 30% and improving platform stability.
- Collaborated with development and operations teams to streamline cloud automation and deployment processes.

Certifications

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| ○ GCP: Professional Cloud DevOps Engineer | ○ Microsoft: Azure Administrator (AZ-104) |
| ○ GCP: Associate Cloud Engineer | ○ Microsoft: Azure DevOps Engineer Expert (AZ-400) |
| ○ HashiCorp: Terraform Associate (003) | ○ Microsoft: Security, Compliance, and Identity Fundamentals (SC-900) |
| ○ Microsoft: Azure Fundamentals (AZ-900) | |

Skills

- **Cloud:** AWS, Azure, GCP — VPC, Networking, Load Balancers, DNS
- **IaC:** Terraform, Terragrunt, Ansible, AWS CloudFormation
- **CI/CD:** Jenkins, GitHub Actions, Azure DevOps, ArgoCD, Atlantis, AWS CodeBuild, Terrateam
- **Containers & Orchestration:** Kubernetes, Docker, Helm
- **Scripting:** Python, Bash, PowerShell, Groovy
- **Monitoring:** Prometheus, Grafana, Loki, CloudWatch, ELK, Splunk,
- **Security:** IAM, RBAC, HashiCorp Vault, Trivy, JFrog Xray, Secrets Management

Education

KLE Technological University

Bachelor of Engineering (B.E.) in Civil Engineering

Hubli, Karnataka

Jun 2017 – Jul 2021